

Raghavan Ramaswamy

raghavansus@gmail.com | (669) 237-8916 | [LinkedIn](#) |

EDUCATION

Carnegie Mellon University | College of Engineering
Masters of Engineering in Mechanical Engineering

Pittsburgh, PA
Expected May 2027

Boston University | College of Engineering
Bachelor of Engineering in Biomedical Engineering

Boston, MA
Graduated May 2024

SKILLS

Hardware Systems: Arduino, Raspberry Pi, LabJack

Hardware Skills: 3D Printing, Lean Eng, DFM, SOLIDWORKS, AUTODESK Fusion, OnShape

Software: Python, Java, C++, C#, MATLAB

EXPERIENCE

Startup Tech Firm(Under NDA)

Boston, MA

Member of Technical Staff

June 2026 - Present

- Designed and built the company's first functional medical-device prototype, integrating mechanical, electrical, and control systems from concept through benchtop validation.
- Completed the Use-Related Risk Analysis (URRA) and Device Description for the company's initial FDA Q-Submission, supporting human factors and regulatory strategy.
- Supporting 510(k) clearance through predicate analysis, V&V planning, risk management, engineering specifications, and design documentation.

IotaMotion

Pittsburgh, PA

R&D Engineering Intern

October 2025 - May 2026

- Designed precision cable-driven mechanisms, reducing mechanical deadzones 3× through iterative structural and routing improvements.
- Characterized cable force transmission at ~80–90% efficiency, identifying up to 25% input-force variation and informing mechanism design.
- Developed a Python automated data-acquisition and analysis pipeline, reducing multi-sensor analysis time by >50%.

RESEARCH

CERLAB @ Carnegie Mellon University

Pittsburgh, PA

Graduate Mechanical Engineering Research Student

September 2025 - Present

- Developed an ESP32-based master-slave endovascular robotic platform with 4 motion-scaling levels and >200 Hz sensor feedback for catheter and guidewire manipulation.
- Designed and conducted a 16-participant human factors study across 4 scaling conditions and 5 targets to quantify accuracy, task performance, and workload; translating findings into a research manuscript.
- Implemented closed-loop motor control achieving <2° steady-state error and developed validation methods that reduced debugging time by ~50%.
- Designed and prototyped a haptic force-feedback system in SOLIDWORKS, characterizing endovascular interaction forces across 100+ trials.

Nia Laboratories

Boston, MA

Biomedical and Mechanical Engineering Researcher

March 2023 - May 2024

- Designed and built a stable 24 hour negative pressure ventilation for mice with a frequency range of 0 - 4 Hz and amplitude range of 0.01 - 0.4 ml with solidworks.
- Spearheaded the end-to-end design and mechanical architecture of a negative pressure ventilator; managed the full product lifecycle from initial CAD requirements to functional prototype within a 2-month rapid-development window.
- Engineered automated test fixtures to simulate 24-hour continuous operation for ventilation systems; conducted Root Cause Analysis (RCA) on motor-encoder failures to restore 100% tracking accuracy.
- Designed custom components in SOLIDWORKS to control a peristaltic pump with a frequency range of 0 - 2 Hz via voltage input, with a focus on DFM (Design for Manufacturing) and assembly constraints for surgical-grade environments.